Computing the subdivision columns on the original graph

Human Compression project: a component of the L059 recovery

Let X be a finite connected simple cubic graph, and let H be an arc-transitive subgroup of its automorphism group. Fix an edge $e = \{u, v\}$ and put

$$P = H_u, \qquad Q = H_e, \qquad B = P \cap Q = H_u \cap H_v.$$

Here H_e fixes the edge as an unordered set. Permutations act on the right, as in the accompanying GAP programs, so that cosets of a subgroup K are written Kh.

The subdivision $S(X)$ has original vertices and edge vertices as its ordered parts, of degrees three and two. Restriction to the original vertices identifies $\mathrm{Aut}(S(X))$ with $\mathrm{Aut}(X)$: degrees distinguish the parts, and the neighbours of each edge vertex determine its two endpoints. Every automorphism of X extends uniquely. Edges of $S(X)$ are incident pairs (u, e), or equivalently arcs (u, v) of X. Thus H gives a faithful marking of $S(X)$ with stabilizers P, Q, B. This is the usual graph/coset correspondence used in De Saedeleer, Leemans, Mixer and Pisanski [1].

The source's four Max entries describe the actions for $B < P$, $B < Q$, $P < H$, $Q < H$. An entry is $2T$ for a doubly transitive action, M for a primitive action that is not doubly transitive, and X for an imprimitive action.

Theorem 1. *The following actual permutation actions are respectively equivalent to the four coset actions defining the Max entries:*

$$P \curvearrowright X(u), \qquad Q \curvearrowright \{u, v\}, \qquad H \curvearrowright V(X), \qquad H \curvearrowright E(X).$$

Replacing the coset actions by these actions changes none of the four entries. It changes neither the acting subgroup classification nor any other invariant column.

Proof. Arc-transitivity of H implies that P is transitive on the three neighbours of u. The stabilizer of v in P is exactly B. Consequently

$$Bp \longmapsto v^p \quad (p \in P)$$

is a bijection from the right cosets of B in P to $X(u)$. It is well defined because two representatives give the same point precisely when their quotient lies in the stabilizer B. It is P-equivariant because right multiplication of a coset by a sends the displayed image to $(v^p)^a = v^{pa}$.

Arc-transitivity also supplies an element of H sending (u, v) to (v, u). It lies in Q, so Q is transitive on the two endpoints of e. The stabilizer of u in Q is B. The same argument gives the Q-equivariant bijection $Bq \mapsto u^q$ onto $\{u, v\}$.

Finally, H is transitive on vertices and on edges. Their stabilizers at u and e are P and Q, respectively, so

$$Ph \longmapsto u^h, \qquad Qh \longmapsto e^h$$

are the other two equivariant bijections. Equivariant bijections carry blocks to blocks and ordered pairs of distinct points to ordered pairs of distinct points. They therefore preserve primitivity and double transitivity, even when an action has a kernel. This proves equality of all four Max entries.

The new routine receives the same list of actual subgroups H, and computes the same P, Q, B. It retains the original normality tests and greatest element order calculation. A subdivision of a cubic graph has no type-exchanging automorphism because its two vertex degrees differ, so its Dual value remains zero. The subgroup enumeration is executed before this routine and is unchanged. Thus neither the classified objects nor the remaining columns are altered. □

Implementation and verification. The separate file `subdivision_actual_actions.g` loads the unchanged original library and replaces only its row routine. It computes each action image explicitly in GAP, then applies the same primitivity and double-transitivity tests. It checks the actual stabilizer intersections, the indices three and two, generation, faithfulness and transitivity.

The independent comparison script uses three already completed census witnesses. It recomputes their rows with the new routine and requires equality of the complete records, including actual subgroups and all invariant columns:

Original cubic input	Marking rows compared
$CAT(1024, 10)$	3
$CAT(1080, 1)$	4
$CAT(1440, 6)$	3

All ten row comparisons pass. This check supplements the uniform equivariance proof; it is not a new enumeration or a completeness assertion about those ten rows.

The change was applied only to 319 future worker scripts. The old worker bytes were retained, and their old and new hashes are recorded in the transition ledger. Both active runners were on the same respective input before and after the change. Four forthcoming inputs for each runner were also left unchanged. The original library and all completed receipts remain unchanged. Future receipts record the new worker bytes; the transition ledger identifies exactly which library those workers load.

The mathematical input is the standard stabilizer/coset equivariance theorem. Our contribution is its application to reuse the existing graph actions in this particular computation. This component makes no new graph classification or whole-original recovery claim. It uses GAP for the finite permutation computations and makes no claim of Lean verification.

References

[1] J. De Saedeleer, D. Leemans, M. Mixer and T. Pisanski, *Core-free, rank two coset geometries from edge-transitive bipartite graphs*, Mathematica Slovaca **64** (2014), 991–1006; arXiv:1106.5704v2.